

TAO (TONY) JIN

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PROFESSIONAL SUMMARY

Quantitative analyst with a Cornell M.Eng. in Systems Engineering and a UBC B.Sc. in Mathematics, plus three internships spanning portfolio risk metrics, financial due diligence, and production data pipelines. Prepares, validates, and aggregates multi-source data, builds risk reporting for senior stakeholders, and translates quantitative results into actionable insights. Strong Excel, SQL, and Python, resolving data-quality issues independently under deadline.

EDUCATION

Cornell University

Master of Engineering in Systems Engineering

Ithaca, NY

Sep. 2025 – May 2026

University of British Columbia

Bachelor of Science: Major in Mathematics

Vancouver, BC

Sep. 2021 – June 2025

- Relevant Coursework: Statistical Inference & Hypothesis Testing, Time Series Analysis, Stochastic Calculus, Optimization, Linear Programming, Relational Databases.

PROFESSIONAL EXPERIENCE

Aifuku

Data Science Intern

Toronto, ON

Jan. 2025 – May 2025

- Prepared, consolidated, and validated operational data across 9 tables and 4 source systems in a Postgres source-of-truth database, writing SQL (joins, CTEs, window functions) to reconcile order, receipt, inventory, and ledger records feeding management dashboards and KPI reporting.
- Assessed data quality across 5 linked entities, independently root-caused reconciliation and schema issues, and presented findings to non-technical stakeholders; built and documented an automated ETL pipeline parsing statement documents into structured records, removing manual data entry.

Grow Investment Group

Institutional Sales Department Intern

Shanghai, China

May 2024 – Aug. 2024

- Completed institutional client due diligence questionnaires (DDQs), cross-referencing multi-source databases with Python (pandas) automation and systematic validation to keep disclosures accurate, cutting manual data processing time 15%.
- Conducted quantitative market research on macroeconomic indicators, interest rate trends, and market sentiment; authored analytical reports and training materials translating findings for non-technical audiences, contributing to 9% fewer redemptions, 4% more inquiries, and 12% higher conversion.

Sinolink Securities Co., Ltd.

Investment Banking Intern

Shanghai, China

June 2023 – Aug. 2023

- Calculated and monitored portfolio risk metrics — Sharpe, Sortino, Beta, volatility, correlation — benchmarking exposures against market indices and consolidating results into risk management reports and presentations for decision-makers.
- Performed pre-IPO financial due diligence (planned raise ~RMB 460M), reconciling multi-year bank statements against ledgers with rigorous attention to detail to escalate unexplained transactions and disclosure discrepancies; standardized data collection and reporting workflows with senior analysts, improving team efficiency 8%.

PROJECT EXPERIENCE

Readmission Risk Modeling & Model Validation | *Python, scikit-learn, XGBoost* Sep.-Dec. 2025

- Built a 160-feature risk model over 101,766 records across 130 sites, benchmarking logistic regression, random forest, gradient boosting, and XGBoost with cross-validation and leakage-safe splitting for out-of-sample validity (AUROC 0.668, AUPRC 0.230).
- Validated and recalibrated the selected model, cutting Expected Calibration Error from 0.342 to 0.004, quantified uncertainty via conformal prediction at 90.1% coverage, and ran driver attribution (SHAP) plus subgroup checks, documented in a written report.

Portfolio Concentration & Fulfillment Risk Analytics (Olist) | *SQL (DuckDB), Excel*

- Aggregated 99,441 transactions into reusable order-, customer-, and item-level data marts, controlling for double-counting at the aggregation grain to protect reported revenue and KPI accuracy; quantified concentration and loss drivers — a BRL 7.98M single-purchase segment, 3.0% repeat rate, 8.1% late deliveries tied to weaker satisfaction — in an executive scorecard.

TECHNICAL SKILLS

Tools & Programming: Microsoft Excel (pivot tables, lookups, dashboards), PowerPoint, Word; SQL (PostgreSQL, Oracle SQL, DuckDB — joins, CTEs, window functions, aggregation); Python (pandas, NumPy, scikit-learn, XGBoost), R, MATLAB, C++, Git, Jupyter

Risk & Quantitative Analytics: Portfolio risk metrics (Sharpe, Sortino, Beta, volatility, correlation), applied statistics, exposure aggregation and concentration analysis, statistical modeling, hypothesis testing, regression, predictive and risk modeling, model validation and calibration, uncertainty quantification, time series forecasting (SARIMA, Holt-Winters), optimization

Data & Reporting: Data preparation, validation, reconciliation, and quality assurance; ETL pipelines; management, compliance, and disclosure reporting; dashboards (Tableau, Power BI, Metabase); KPI reporting; data visualization; ad hoc analysis

Business & Communication: Excellent written, verbal, and presentation skills; stakeholder communication; cross-functional collaboration; translating complex analysis into actionable insights for senior leadership; prioritizing multiple assignments in a fast-paced environment; critical thinking; attention to detail. Languages: Chinese (Native), English (Professional), Japanese (Basic)